EE561 Advanced Static Power Conversion

Multilevel Converters: Building the Levels — Quiz Answer Key

Sefer İdacı · for the presenter and the instructor · not for distribution to students

Question 1 · Device selection — 25 pts

(a) Required rating = blocked voltage ÷ (2/3) = 1.5 × blocked voltage; pick the next class.

Leg / device	Blocked voltage	Needed rating	Class
2-level, each switch	1100 V	1650 V	**1700 V**
3L NPC, each switch	550 V	825 V	**1200 V**
3L T-type, outer S1, S4	1100 V	1650 V	**1700 V**
3L T-type, inner S2, S3	550 V	825 V	**1200 V**

(b) Advantage: in P and N only one device conducts (NPC: two) → lower conduction loss; no clamp diodes. Disadvantage: the outer switches must block the full 1100 V → 1700 V class: slower, more switching loss, fewer and costlier parts.

Marking: (a) 3 pts per row + 3 pts for the 2/3 rule; (b) 5 + 5.

Question 2 · NPC leg in the zero state — 25 pts

(a) Path: output a → **S3** → node between S3 and S4 → **D6** → midpoint O. Conducting devices: S3 and D6.

(b) Current flows from the output into O: the leg **pushes current into the midpoint** (current drawn from O is $i_O = i_a < 0$).

(c) S1 blocks 400 V (P to node X, which sits at O through S2) and **S4 blocks 400 V** (node Y at O to N). S2, S3, D5, D6 see ≈ 0 V.

(d) Going P → N directly switches both complementary pairs at once: the output jumps by the full 800 V (twice the dv/dt and EMI of a normal step), and during the transient the series devices S1–S2 or S3–S4 must share 800 V with no clamp guaranteeing equal sharing → one device can be overvoltaged. Nothing is saved: switching loss scales with the voltage step. Always pass through O.

Marking: (a) path 4 + devices 3; (b) 5; (c) 3 + 3; (d) step size 3, voltage sharing 3, conclusion 1.